ECOTEK – Controller User Guide & Error Codes

Comprehensive Customer Interface Reference, Key Combinations & Diagnostic Trouble Shooting

CUSTOMER QUICK START NOTICE
This guide provides complete, customer-accessible instructions for operating ECOTEK R32 Air Source Heat Pump wired controllers (LCD & TFT Touchscreen series). Use this document to master screen functions, lock/unlock sequences, mode changes, temperature adjustments, weekly timer programming, and diagnostic troubleshooting.

1. Controller Interface & Button Functions

ECOTEK heat pump systems feature advanced wired controllers designed for intuitive residential and commercial climate management. The controller display provides real-time operational status, temperature readouts, active modes, and diagnostic indicators.

Button / Icon	Name	Primary Function & Operation Details
Power (ON/OFF)	**Power / Lock Key**	Turns heat pump ON or OFF (press & hold 1–2s when unlocked). Acts as Keypad Unlock when locked (press & hold 3–5s until lock icon disappears).
Mode (M)	**Mode Selection**	Switches operating modes: Heating, Cooling, Domestic Hot Water (DHW), and Combined Modes. Confirms settings in programming menus.
Clock / Timer ()	**Clock & Timer Key**	Short press to adjust time of day. Press & hold 3s to enable/disable Weekly Timer mode or clear daily timers.
Up Arrow ()	**Increase / Scroll**	Raises setpoint temperature, increases numerical values in parameter menus, and scrolls up through query status codes.
Down Arrow ()	**Decrease / Status**	Lowers setpoint temperature, decreases values in parameter menus, and enters Operating Status Query mode (press & hold 3s).

Display Screen Icon Directory

Icon Graphic / Label	State / Mode	Functional Meaning & System Indication
Lock ()	**Solid**	Keypad is locked. Press and hold Power button for 3–5 seconds to unlock.
Sun / Flame (/)	**Solid / Flashing**	Heating Mode active. Solid = operating normally; Flashing = unit demand active.
Snowflake ()	**Solid / Flashing**	Cooling Mode active. Provides chilled water for fan coils or space cooling.
Water Tap / Tank (/)	**Solid / Flashing**	Domestic Hot Water (DHW) Mode active. System prioritizes hot water tank heating.
Floor Grid ()	**Solid / Flashing**	Underfloor Heating Mode active. Optimized low-temperature heating supply.
Moon / Leaf (/)	**Solid**	Silent Mode (Quiet Operation) active. Compressor and fan speeds are capped.
Lightning / Power ()	**Solid**	Powerful / Booster Mode active. Maximum heating/cooling ramp-up speed.
Sterilization (/70)	**Solid / Flashing**	High-Temperature Anti-Legionella Sterilization active (DHW tank heated to 70°C).
Wi-Fi Signal ()	**Fast / Slow Flash**	Fast flash = Smart Distribution Mode pairing; Slow flash = AP Hotspot pairing; Solid = Online.
Water Pump ()	**Solid / Flashing**	Solid = Circulation pump running; Flashing = Anti-freeze freeze-proof pump operation.
Compressor ()	**Solid**	Outdoor unit compressor is running.
Defrost ()	**Flashing**	Automatic Outdoor Coil Defrosting in progress (fan stops, 2–8 minutes).
Warning Triangle (/ Exx)	**Flashing**	Fault Alarm triggered. System displays alternating Error Code (e.g., E03, E37).

2. Detailed Button Combinations & Procedure Guide

Follow these exact step-by-step procedures for controlling and configuring your ECOTEK heat pump. Ensure the controller is unlocked before attempting key sequences.

Unlocking & Locking Keypad

Unlock: Press and hold the **Power/Unlock** button for 3 to 5 seconds until the Lock () icon disappears.
Lock: Automatic after 60 seconds of key inactivity (or press & hold Power button for 3s while unlocked).

Powering ON / OFF the System

1. Ensure keypad is unlocked.
2. Press and hold the **Power (ON/OFF)** button for **1 to 2 seconds**.
3. The display illuminates and shows operational status (ON) or enters standby mode (OFF).

Changing Operating Modes

1. With screen unlocked and powered ON, press the **Mode (M)** button once.
2. Repeatedly press **Mode (M)** to cycle through available modes:
• **Heating:** Space heating only.
• **Cooling:** Space cooling (fan coil/chilled water) only.
• **DHW (Hot Water):** Domestic hot water tank only.
• **Heating + DHW:** Combined mode (DHW has priority via 3-way valve).
• **Cooling + DHW:** Combined cooling and sanitary hot water.
• **Underfloor Heating:** Dedicated low-temperature floor heating.

Adjusting Target Temperatures

1. From homepage (unlocked), press **Up ()** or **Down ()** once. Target temperature digits flash.
2. Press **Up ()** to raise or **Down ()** to lower desired setpoint.
3. In **Combined Modes** (e.g., Heating + DHW), short press **Mode (M)** while flashing to switch setting between Space Temp and Tank Temp.
4. Wait 5 seconds without pressing any key (or press Power ON/OFF) to save and return to homepage.

Setting Real-Time Clock (Time of Day)

1. Unlock screen. Press **Clock / Timer ()** for 1 second. Hour digits flash.
2. Use **Up () / Down ()** to set current Hour (00–23).
3. Press **Clock ()** again. Minute digits flash.
4. Use **Up () / Down ()** to set current Minute (00–59).
5. Press **Clock ()** again or wait 5 seconds to save settings.

Weekly Timer & Schedule Management

Enable / Disable Weekly Timer: Unlock screen, press **Clock ()** once so time flashes, then press and hold **Clock () for 3 seconds**. 'W' icon turns ON or OFF.
Programming Timers (Group 1–3):
1. Press & hold **Clock () for 3–5 seconds** to enter Timer Setup.
2. If Weekly mode is ON, select Day of Week (W1–W7 / Mon–Sun) using **/**, then press **Clock**.
3. '1 ON' flashes: Set Start Hour & Minute using **/**, press **Clock** to confirm.
4. '1 OFF' flashes: Set Stop Hour & Minute using **/**, press **Clock** to confirm.
5. Repeat for Groups 2 and 3. Wait 5 seconds to auto-save.

ECO Energy-Saving Mode

Standard Controller: In unlocked state, press & hold **Up () + Down ()** simultaneously for 3 seconds. The **ECO** symbol illuminates.
TB Series Controller: Long press **Up ()** or **Down ()** for 3 seconds on main screen.
Function: Automatically applies energy-optimized temperature curves (Curve 4/6).

Quick Heating / Booster Mode

In unlocked state (power ON), press & hold **Mode (M) + Up ()** (or **Power + Up**) simultaneously for 3 seconds.
The Quick Heat symbol () lights up, energizing electric backup heaters (EH1/EH2) for rapid temperature recovery.

Silent / Night Mode

In unlocked state, press & hold **Timer () + Down ()** (or **Mode + Timer**) for 3 seconds.
The Silent Mode symbol () turns ON. Caps compressor frequency to 50Hz and fan speed to 40Hz to minimize outdoor noise.

Water Line Evacuation / Forced Pump Mode

In **Power OFF** state, press & hold **Power + Up () + Down ()** simultaneously for 5 seconds.
The Pump symbol () flashes and the circulation pump runs continuously for water system purging/air bleeding. Press Power to exit.

Wi-Fi Smart Life App Pairing

Smart Mode (Fast Flashing): Unlocked state Press & hold **Power + Up ()** for 3s. Open Smart Life App Add Device Auto-detect Heat Pump.
AP Mode (Slow Flashing): Unlocked state Press & hold **Power + Down ()** for 3s. Connect phone to hotspot 'smartlife-XXXX' Complete binding in App.

Status Parameter Query (Live Data Readout)

1. In unlocked state, press & hold **Down () for 3 to 5 seconds**.
2. Temperature area displays **Query Code (1–30)**; Timer area displays live value.
3. Press **Up () / Down ()** to scroll through parameters:
• **Code 1:** Compressor Freq (Hz) | **Code 2:** Fan Speed (Hz) | **Code 11:** Outdoor Ambient Temp (°C)
• **Code 16:** Water Inlet Temp (°C) | **Code 17:** Water Outlet Temp (°C) | **Code 21:** Tank Temp (°C)
• **Code 25:** Water Flow Rate (L/min) | **Code 26:** Return Water Temp (°C) | **Code 30:** Total kWh.
4. Press **Power (ON/OFF)** to exit Query Mode.

3. Comprehensive Error Codes & Customer Troubleshooting

When an operational anomaly occurs, the controller display flashes a yellow warning icon and cycles the corresponding **Error Code (E01–E99, EA1–EA5, EC0)**. Review the complete diagnostic reference below for root causes and safe customer corrective steps.

Code	Error Description	Root Cause Analysis	Customer Troubleshooting & Corrective Action
E03	**Outside Water Flow Switch Fault**	No water flow, closed valves, pump failure, air locked in pipes, clogged filter, or faulty flow switch.	1. Verify circulation pump is running. 2. Open all water valves. 3. Clean water filter mesh. 4. Purge trapped air using automatic/manual air vents.
E32	**Excessive Water Temp / Overheating**	Water flow too low, thermal overload, or outlet sensor fault.	1. Check water flow rate. 2. Ensure circulating pump is operating. 3. Power off for 5 mins and restart.
E37	**Large Temp Difference Inlet/Outlet**	Extreme temperature gap between Inlet (T8) and Outlet (T15) due to restricted water flow or airlock.	1. Inspect water circulating pump. 2. Clean inline filter. 3. Fully purge air from heating loop. 4. Check sensor thermowell seating.
EC0	**Low Water Flow Protection (Fault 120)**	Measured water flow rate dropped below minimum protection limit (Parameter P134).	1. Check for valve restrictions. 2. Verify water pump speed setting. 3. Inspect filter strainer for debris.
E14	**Hot Water Tank Temp Sensor Fault**	Sensor T16 (Blue lead) disconnected, shorted, or mainboard port damaged.	Check T16 sensor cable connection to tank thermowell. If disconnected, reconnect. Contact service if persistent.
E15	**Water Inlet Temp Sensor Fault**	Inlet Sensor T8 cable damaged, loose, or failed.	Inspect T8 sensor lead on inlet piping. Restart controller.
E16	**Coil Sensor 1 Failure**	Outdoor coil sensor T1 damaged or loose.	Contact service technician to check sensor resistance.
E17	**Coil Sensor 2 Failure**	Secondary coil sensor T4 damaged or loose.	Contact service technician.
E18	**Exhaust Gas Temp Sensor 1 Fault**	Exhaust sensor T3 wire damaged or open circuit.	Contact service technician.

E20	**Indoor Temp Sensor Failure**	Indoor room sensor damaged or disconnected.	Check room thermostat / indoor sensor cable.
E21	**Outdoor Ambient Temp Sensor Fault**	Ambient sensor T7 damaged or reading out of range.	Ensure outdoor unit sensor is clean and unblocked by ice/snow.
E22	**User Return Water Sensor Failure**	Return water sensor T13 (Red lead) faulty.	Inspect T13 sensor lead on return pipe loop.
E23	**Cooling Subcooling / Low Outlet Temp**	Water outlet temperature dropped too low during cooling mode (anti-freeze risk).	Increase cooling water setpoint. Ensure sufficient water flow through fan coils.
E24	**Anti-Freeze Sensor Fault (Refrigerant)**	Refrigerant circuit freeze sensor T14 faulty.	Contact service technician.
E26	**Anti-Freeze Sensor Fault (Water Side)**	Water circuit freeze protection sensor faulty.	Contact service technician.
E27	**Water Outlet Temp Sensor Failure**	Outlet sensor T15 damaged or disconnected.	Inspect T15 sensor lead on discharge pipe.
E29	**Suction Temp Sensor 1 Fault**	Compressor suction sensor T2 damaged.	Contact service technician.
E01	**Wrong Phase Protection**	3-Phase electrical power phase sequence inverted.	Technician must swap two incoming phase wires (R-S-T) at breaker.
E02	**Missing Phase Protection**	3-Phase electrical supply missing one phase.	Check main power breaker and incoming voltage lines.
E05	**High Pressure Switch 1 Fault**	High refrigerant pressure due to fan failure, dirty heat exchanger, excess gas, or low water flow.	1. Verify outdoor fan is spinning. 2. Ensure water pump is circulating. 3. Clean outdoor coil fins.
E06	**Low Pressure Switch 1 Fault**	Low refrigerant pressure due to leak, severe cold ambient, or ice blockage on evaporator.	1. Check for severe ice accumulation on outdoor coil. 2. Turn off unit and call service if refrigerant leak suspected.
E07	**High Pressure Switch 2 Fault**	System 2 high pressure protection.	Same as E05 for dual compressor units.
E08	**Low Pressure Switch 2 Fault**	System 2 low pressure protection.	Same as E06 for dual compressor units.
E09	**Controller Communication Failure**	Communication signal lost between wired controller and mainboard (COM4 port).	1. Turn off power breaker. 2. Inspect 4-wire signal cable for loose screws or breaks. 3. Restore power after 60s.
E11	**Limited Time / Trial Lock Protection**	Service trial period expired or security lock.	Contact installing dealer to enter activation code.
E12	**Exhaust Gas Temp 1 Too High**	Compressor discharge temperature exceeded 110°C due to gas shortage or blockage.	Shut down unit immediately. Contact service technician.
E31	**Water Pressure Switch Failure**	Water loop static pressure dropped below 0.15 MPa (1.5 bar).	Refill water system via feed valve until gauge reads 0.15–0.20 MPa (1.5–2.0 bar).
E38	**DC Fan 1 Failure**	Outdoor fan motor 1 or fan driver board AP5 faulty.	Inspect fan blade for debris obstruction. Call service.
E39	**DC Fan 2 Failure**	Outdoor fan motor 2 faulty.	Inspect fan blade for obstruction.
E44	**Low Ambient Temp Protection**	Outdoor ambient temperature below -30°C operating limit.	Standard safety cutout. Unit automatically restarts when ambient temperature rises.
E51	**High Pressure 1 Overvoltage**	System 1 overpressure protection.	Same as E05.
E52	**Low Pressure 1 Undervoltage**	System 1 underpressure protection.	Same as E06.
E55	**Expansion Board Communication Error**	Signal connection failure between mainboard AP1 and expansion board AP3.	Turn off power, inspect board interconnect ribbon cable.
E80	**Power Supply Error**	Single-phase unit connected to 3-phase power line or miswired.	Check incoming electrical voltage specification.
E88	**Inverter Module 1 Protection (P1–P43)**	Compressor inverter drive board AP2 protection (Overcurrent, IPM overheat, bus voltage).	1. Disconnect main breaker for 3 minutes to reset inverter. 2. If error recurs, contact qualified service technician.

E89	**Inverter Module 2 Protection**	Compressor inverter drive 2 protection.	Same as E88 for System 2.
E94	**Water Pump Feedback / Voltage Fault**	Built-in DC pump supply voltage <165V or >265V, or internal pump board failure.	Check main supply voltage stability. Inspect pump signal connector.
E96	**Compressor Driver 1 Comm Failure**	Communication lost between mainboard AP1 and compressor driver AP2.	Turn off power breaker for 60s and re-power.
E98	**Fan Motor Driver Comm Failure**	Communication lost between mainboard AP1 and fan driver AP5.	Turn off power breaker for 60s and re-power.
EA1	**Cascade Model Mismatch (101)**	Incompatible heat pump models connected on cascade RS485 bus.	Ensure all cascaded units belong to identical model series.
EA2	**Solar Water Sensor Fault (102)**	Solar temperature sensor T12 lead damaged.	Inspect T12 sensor lead.
EA3	**Zone 2 Temp Sensor Fault (103)**	Dual-zone 2 temperature sensor T11 lead damaged.	Inspect T11 sensor lead.
EA4	**Buffer Tank Sensor Fault (104)**	Buffer tank temperature sensor T10 (Green lead) damaged.	Inspect T10 sensor cable and thermowell.
EA5	**Total Outlet Sensor Fault (105)**	Total water outlet sensor T9 (Black lead) damaged.	Inspect T9 sensor lead on discharge header.

IMPORTANT FAULT RESET & SAFETY WARNING

Standard System Reset Protocol: To clear persistent fault lockouts after rectifying the root cause (e.g., refilling water or clearing air), turn OFF the main electrical supply circuit breaker to the outdoor heat pump for **at least 60 seconds** to drain power from the control board memory, then restore power.

Safety Warning: If Error E05, E12, E88, or E89 recurs immediately after resetting, do not attempt further power cycles. Turn off the main breaker and contact a certified HVAC technician.